Part E - written justification

Esha Asif - Roll 034 - BSAI 7A - CDB601220 Week 4. Synthetic course data throughout; nothing here is a finding about a real business.

1. The four panels: the question, the chart, the channel rule

Panel 1 (daily sessions, line): where is traffic heading once the weekend rhythm is out? A line puts value on *position along a common scale*, the top of the Week 2 channel ranking (position > length > angle > area); the raw series sits in grey as context and the 7-day mean is the single pop-out. **Panel 2 (day x hour heatmap):** when in the week do people visit? Colour lightness (one hue, sequential, no midpoint) is a weaker channel, so it is used for a pattern to be seen as a region, with the colour bar carrying the scale. **Panel 3 (treemap):** where is the mass of revenue? Area is the fourth-best channel (about +/-20%), so every block that fits carries its value, and hue marks the parent. **Panel 4 (sorted bars):** by how much? Length on a shared baseline is read far more accurately than area; values are printed. Panels 3 and 4 are the same eight numbers: the treemap shows where the mass is, the bars show by how much.

2. Every time decision, named and defended

Parse. The file mixes two formats: 9,019 ISO rows and 3,651 dd/mm/yyyy rows. The slash rows are day-first, and I proved it rather than assumed it: 2,151 of them have a first field above 12 and 0 have a second field above 12. Each format is parsed with its own explicit format string. *format='mixed'* would also parse every row without an error, but it moves 1,382 rows to the wrong day and stretches the series to 2024-01-04 - 2025-12-08 instead of the true 2024-04-01 - 2025-09-11. **Aggregation.** Sessions are a flow, so the hourly rows are summed to days (sum, never last). The 13 unrecorded days are counted on the *hourly* index before resampling (312 hours, all 24 of each day), left as NaN and the line breaks there; nothing is interpolated in the figure. **Window.** 7 days, centred: the cycle being cancelled is weekly, and a window one cycle long holds every weekday once (Part B.1: it removes 31.8% of the raw variation, a 91-day window only 39.1%). Centred, because this is a retrospective Monday review; the first and last 3 days have no smoothed value, and a live version would need the trailing mean and its 3-day lag. **Partial month.** 2025-09 holds 11 of 30 days; it is kept and marked on panel 1. The year-on-year figure in the title compares Apr-Aug only, the months complete in both years.

The window clause. The window was fixed at 7 days before plotting, because the cycle being removed is weekly (Part B.1). The series is shown from its first recorded day, 1 April 2024; no start date was selected. A 28-day version is in the appendix (partD_appendix_28day.png) and does not change the direction: the fitted slope is +21.1 sessions/day per day at 7 days and +19.7 at 28, and the Apr-Aug year-on-year change on matched dates is +45.2% against +45.9% (the 28-day line has far fewer paired days, because gaps break a wide window). Part B.5 shows how much the choice could matter: of 60 start-and-window combinations, 32 rise and 28 fall.

Symmetry test: would I have made the same choices if they had weakened my conclusion? Yes - the window comes from the weekly cycle, the start is the first row of the file, and the appendix is there because the 28-day line could have disagreed.

3. Every row removed: rule, count, and what the number moved from and to

Fault	Rule	Count	Moved from -> to
Duplicate hours	keep first (all identical)	286 rows	rows 12,670 -> 12,384
Sentinel -1 sessions	set to NaN, not 0	290 raw / 287 after de-dup, on 222 days	mean sessions/hour 946.94 -> 969.43
Device spellings	strip + lower-case	6 spellings -> 3	rows unchanged
Unrecorded days	not filled; line breaks	13 days = 312 hours	days with data 529 -> 516
Duplicate SKU-0127	keep Audio copy (name AUDI-04)	1 row	Wearables share 10.14% -> 9.36%; largest gap to shares_eight.csv 0.23 pts (was ~0.9)
Negative net SKUs	not drawn; footnoted on the figure	7 SKUs, PKR -1.07m	net total PKR 1,711.38m -> 1,712.45m (+1 SKU with net exactly 0)

Black Friday (113,710 sessions) is a real event, not a fault: it stays in. Days with one to three dropped hours (222) read slightly low because the day is summed from the hours that were recorded; they are flagged, not filled.

4. Every level rolled up

The catalogue has 336 drawable SKUs. They are rolled up to **8 sub-categories under 4 categories**, and I cut the depth there because 336 SKU blocks are unreadable and the chosen level answers the question. There is **no 'Other' bucket**: all 8 sub-categories are drawn, the smallest (Drones, 2.4%) carries a short label in the treemap and its exact value on panel 4. The fourth hue (pink) always carries a printed label, as vizlib requires. The sub-category shares match shares_eight.csv to within 0.23 percentage points, so the pie of Part C and this treemap are one whole at two grains.